

Question 1. (2 + 2 + 2 + 2 marks) Prove or disprove the following statements. Disproving may be done via counter-examples.

- (a) If two states in a Markov Chain do not communicate, then they cannot have the same period.
- (b) Let P denote the transition probability matrix on a finite state Markov Chain. Then $P^n, n = 2, 3, \dots$ are stochastic matrices.
- (c) A Markov chain on a countably infinite state space necessarily has all states being recurrent.
- (d) Communication classes in a Markov chain are closed subsets of the state space.

Hints:

- (a) Consider any Markov Chain on the state space $\{0, 1, 2, 3\}$ and TP matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Check that $\{0, 1\}$ and $\{2, 3\}$ are Communication classes. Periods of 0 and 2 are both 2, even though they do not communicate. The statement is false.

- (b) Using the principle of mathematical induction show that the entries of P^n are non-negative. For the row-sum, observe that

$$P \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

Using the principle of mathematical induction show that

$$P^n \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

The statement is true.

- (c) (discussed in class) Asymmetric Random walk in dimension 1 is a Markov chain on a countably infinite state space and is transient. The statement is not true.
- (d) (discussed in class) In the following two-person zero-sum game with state space $\{0, 1, 2, 3\}$

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$\{1, 2\}$ is a Communication class. But, as $p_{10} = \frac{1}{2} > 0$, $1 \rightarrow 0$ and $\{1, 2\}$ is not closed. The statement is not true.